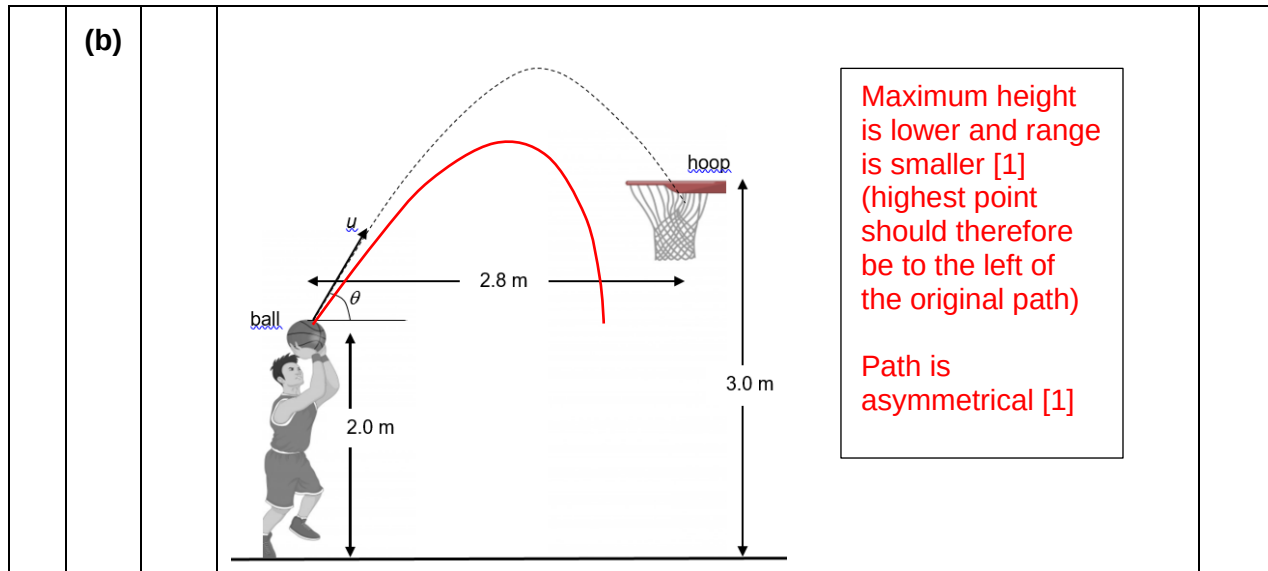


			Thus, mass of the astronaut = (65 ± 2) kg	1
2	(a)	(i)	$s_y = u_y t + \frac{1}{2} a_y t^2$ $\Rightarrow 1.0 = u_y (1.1) + \frac{1}{2} (-9.81)(1.1)^2$ $\Rightarrow u_y = 6.30 \text{ m s}^{-1}$	1 1
		(ii)	$s_x = u_x t$ $\Rightarrow 2.8 = u_x (1.1)$ $\Rightarrow u_x = 2.545$ Thus, $\theta = \tan^{-1}\left(\frac{u_y}{u_x}\right)$ $= \tan^{-1}\left(\frac{6.305}{2.545}\right) = 68.0^\circ$	1 1 1



3	(c)	The spring does not obey Hooke's Law	1
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